

Notes: Bollen and Busse (RFS, 2005)

Short-Term Persistence in Mutual Fund Performance

Contents

1	Big Picture	2
1.1	Research question	2
1.2	Core answer	2
1.3	Main intuition	2
2	Incremental Contribution	2
2.1	What the paper adds relative to prior work	2
2.2	Why it matters	3
3	Data	3
3.1	Mutual fund sample	3
3.2	Important sample feature	3
3.3	Factor data	3
4	Performance Models	3
4.1	Stock selection model	3
4.2	Market timing models	4
4.3	Timing performance measure	4
4.4	Mixed strategy model	4
5	Empirical Design	4
5.1	Quarterly ranking and post-ranking tests	4
5.2	Why daily data are essential	4
5.3	Persistence regression	4
6	Main Findings	5
6.1	Finding 1: Ranking-quarter dispersion is large	5
6.2	Finding 2: The top decile continues to perform in the next quarter	5
6.3	Finding 3: Figure 1 shows the short-term pattern directly	5
6.4	Finding 4: Raw returns hide part of the persistence	5
6.5	Finding 5: Cross-sectional persistence is significant for abnormal performance, not raw returns	5
6.6	Finding 6: Robustness tests do not support a simple momentum or microstructure explanation	5
6.7	Finding 7: Persistence is short-lived	6

7	Interpretation	6
7.1	Short-term ability versus long-term unconditional performance	6
7.2	Why Carhart-style evidence differs	6
7.3	Role of fund flows and factor loadings	6
7.4	Economic significance	6
8	Limitations and Caveats	6
9	Detailed Table and Figure Walkthrough	7
9.1	Table 1: Risk-adjusted performance deciles: Ranking quarter	7
9.2	Table 2: Risk-adjusted performance deciles: Post-ranking quarter	7
9.3	Figure 1: Cumulative abnormal returns	8
9.4	Table 3: Post-ranking period performance deciles: Total returns	8
9.5	Table 4: Post-ranking period performance deciles: Sharpe ratios	9
9.6	Table 5: Factor loadings of risk-adjusted performance deciles	9
9.7	Table 6: Cross-sectional regression tests of performance persistence	10
9.8	Table 7: Specification test	10
9.9	Table 8: Robustness test	11
9.10	Table 9: Top decile post-ranking period performance by length of ranking and post-ranking period: Time series averages – sorting on abnormal return	11
9.11	Table 10: Top decile post-ranking period performance by length of ranking and post-ranking period: Time series averages – sorting on return	12
9.12	Table 11: Top decile post-ranking period performance by length of ranking and post-ranking period: Concatenated series – sorting on abnormal return	12
9.13	Table 12: Top decile post-ranking period performance by length of ranking and post-ranking period: Concatenated series – sorting on return	13
10	Bottom Line	14

1 Big Picture

1.1 Research question

The paper asks whether mutual fund performance persists over a short horizon. Earlier studies usually use monthly returns and long evaluation windows. Bollen and Busse ask whether those choices miss a short-lived form of managerial ability that can only be detected when funds are evaluated several times per year.

1.2 Core answer

The paper finds statistically significant short-term persistence. Funds ranked in the top decile by risk-adjusted performance over one quarter continue to earn positive abnormal returns in the next quarter. Across stock selection, market timing, and mixed strategy models, the post-ranking abnormal return for the top decile is roughly 25–39 basis points per quarter. The effect disappears when the authors use longer measurement periods, rank on raw returns rather than abnormal returns, or evaluate performance using a single concatenated long time series.

1.3 Main intuition

The key idea is that skill, information, or favorable opportunities may decay quickly. If investors chase recent winners and capital flows into high-performing funds, the advantage of those funds may be competed away. Therefore, a one-year or three-year measurement window may average away a signal that is visible over only a quarter.

2 Incremental Contribution

2.1 What the paper adds relative to prior work

1. **Moves the performance horizon from years to quarters.** Prior persistence studies typically use monthly data and one-year or longer ranking windows. This paper uses daily mutual fund returns to estimate risk-adjusted performance within three-month windows.
2. **Studies both stock selection and market timing persistence.** The paper does not only rank funds by the Carhart-style intercept; it also estimates market timing models and asks whether timing performance persists.
3. **Allows strategy heterogeneity and strategy switching.** The mixed strategy tests recognize that some managers may behave like stock pickers, others like market timers, and a given manager may change strategy over time.
4. **Explains why results differ from Carhart (1997).** The paper shows that sorting on raw returns, using long measurement windows, and estimating performance on concatenated returns all remove the abnormal return. These design choices help reconcile the paper with the long-horizon no-skill evidence.
5. **Separates statistical from economic significance.** The paper is careful that short-run predictability may not be exploitable after transaction costs and taxes.

2.2 Why it matters

The result changes the interpretation of mutual fund skill. A long-horizon null result does not necessarily mean no manager ever has ability. It may mean that ability is transient, difficult to isolate with low-frequency returns, and hard for investors to exploit net of trading frictions. The paper therefore pushes mutual fund performance analysis toward higher-frequency data, shorter windows, and conditional evaluation.

3 Data

3.1 Mutual fund sample

The sample contains 230 domestic equity mutual funds from January 2, 1985 to December 29, 1995. Funds are selected from the December 1984 Wiesenberger universe with common stock investment policy and growth-oriented objectives. Sector, balanced, and index funds are excluded. Daily net asset values and dividends are used to construct daily fund returns.

3.2 Important sample feature

The sample avoids the usual form of survivorship bias because funds that disappear during the sample are not dropped. However, it does not include funds that are born after the December 1984 selection date. Thus, the sample is well suited for studying persistence among the existing equity fund universe but not for studying fund entry.

3.3 Factor data

The performance models use daily versions of the market excess return, the Fama–French size and book-to-market factors, and a Carhart momentum factor. Lagged values of factors are also included in the estimation to reduce problems from infrequent trading and stale prices in daily returns.

4 Performance Models

4.1 Stock selection model

The stock selection test uses the intercept from a four-factor model as the measure of abnormal performance:

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{p,k} r_{k,t} + \varepsilon_{p,t}.$$

Here $r_{p,t}$ is the excess return on fund p and $r_{k,t}$ are the four factor returns. A positive α_p is interpreted as abnormal return from stock selection after controlling for market, size, value, and momentum exposure.

4.2 Market timing models

The paper also estimates market timing models of the form

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{p,k} r_{k,t} + \gamma_p f(r_{m,t}) + \varepsilon_{p,t}.$$

For the Treynor–Mazuy model, $f(r_{m,t}) = r_{m,t}^2$. For the Henriksson–Merton model, $f(r_{m,t}) = I_t r_{m,t}$, where the indicator captures high-market-return states. A positive timing coefficient means the fund return is convex in the market return, as would occur if the manager raises market exposure before good market states.

4.3 Timing performance measure

For timing models, the paper ranks funds using the average contribution of timing to daily abnormal return:

$$r_{p,\gamma} = \frac{1}{N} \sum_{t=1}^N [\alpha_p + \gamma_p f(r_{m,t})],$$

where N is the number of trading days in the quarter.

4.4 Mixed strategy model

The mixed strategy sort first classifies a fund as a market timer in a quarter if its timing coefficient is statistically significant. If it is a timer, the fund is ranked by the timing performance measure. If not, it is ranked by the stock-selection intercept. This design recognizes that not every fund is trying to create value in the same way.

5 Empirical Design

5.1 Quarterly ranking and post-ranking tests

For each quarter except the last, the authors estimate fund performance over the ranking quarter. They rank funds into deciles and then measure the performance of each decile during the following quarter. The central test is whether the top decile in the ranking quarter continues to generate abnormal returns in the post-ranking quarter.

5.2 Why daily data are essential

A three-month window contains too few monthly observations to estimate a factor model precisely. Daily returns create enough observations to estimate risk-adjusted performance over a quarter. The paper’s empirical contribution depends on this high-frequency measurement.

5.3 Persistence regression

The authors also estimate cross-sectional regressions of performance on lagged performance:

$$Perf_{p,t} = a + b Perf_{p,t-1} + \varepsilon_{p,t}.$$

A positive and significant b means that past quarterly performance predicts next-quarter performance. The authors estimate the regression quarter by quarter and use Fama–MacBeth-style inference.

6 Main Findings

6.1 Finding 1: Ranking-quarter dispersion is large

The ranking-quarter spread is mechanically large because the funds are sorted on estimated performance. Under the stock selection model, the top decile has a daily abnormal return of 0.0738%, while the bottom decile has -0.0828%. The ranking-period spread shows that the daily model identifies substantial cross-sectional dispersion in estimated performance.

6.2 Finding 2: The top decile continues to perform in the next quarter

In the post-ranking quarter, the top stock-selection decile earns 0.0061% per day, about 39 basis points per quarter. The bottom decile earns -0.0122% per day, about -77 basis points per quarter. The timing and mixed models produce similar qualitative conclusions, with the top decile earning positive post-ranking abnormal returns.

6.3 Finding 3: Figure 1 shows the short-term pattern directly

The figure plots cumulative post-ranking abnormal returns for the top and bottom stock-selection deciles. The top decile earns most of its abnormal return in the first half of the quarter, while the bottom decile declines relatively steadily. The steady poor performance of the bottom decile is consistent with persistent expenses or bad performance, a pattern also emphasized by Carhart.

6.4 Finding 4: Raw returns hide part of the persistence

Post-ranking raw returns and Sharpe ratios differ much less across deciles than risk-adjusted alphas. The reason is that the top and bottom deciles have different factor loadings. For example, the top decile has a higher exposure to size and momentum and a more negative loading on the value factor. Risk adjustment is therefore central to seeing the persistence result.

6.5 Finding 5: Cross-sectional persistence is significant for abnormal performance, not raw returns

Table 6 shows that the slope coefficient in the persistence regression is positive and significant for the risk-adjusted performance measures. It is insignificant when performance is measured by raw returns. This reinforces the message that ranking on raw returns and ranking on estimated abnormal returns identify different funds.

6.6 Finding 6: Robustness tests do not support a simple momentum or microstructure explanation

The authors simulate size and momentum strategies, subtract reasonable fees and transaction costs, and show that the adjusted simulated abnormal returns are usually negative. They also create

alternative momentum factors at horizons from 12 months to one day. The top decile remains positive across these alternative specifications. Other diagnostics address cross-autocorrelation and microstructure concerns.

6.7 Finding 7: Persistence is short-lived

When performance is measured over one-year or three-year windows, the top-decile abnormal return becomes small and statistically insignificant. When post-ranking returns are concatenated into one long time series and performance is estimated once, the evidence of ability disappears. This finding is the main reconciliation with Carhart’s long-horizon results.

7 Interpretation

7.1 Short-term ability versus long-term unconditional performance

The paper’s main interpretation is that superior performance is conditional and short-lived. A fund can look skilled over a short horizon without producing a long-horizon unconditional alpha. The long-horizon performance measure mixes changing factor loadings, flows, strategy changes, and changing opportunity sets.

7.2 Why Carhart-style evidence differs

Carhart’s analysis differs in three major ways. First, it ranks funds by raw returns or long-run abnormal performance rather than by quarterly abnormal return. Second, it uses longer measurement windows. Third, it often uses concatenated post-ranking returns. Bollen and Busse show that each of these changes pushes their results toward the Carhart no-skill conclusion.

7.3 Role of fund flows and factor loadings

The paper links the gap between short- and long-run evidence to the difference between conditional and unconditional performance. Fund flows can change funds’ factor loadings over time. Quarterly estimation allows factor loadings to vary; a single long time-series regression imposes much more stability.

7.4 Economic significance

The short-run abnormal return is statistically meaningful but not necessarily easy to monetize. An investor chasing the top decile every quarter would face transaction costs, taxes, and possible delays in observing performance. Thus, the paper’s evidence is stronger as a statement about temporary predictability than as a simple trading recommendation.

8 Limitations and Caveats

- **Sample construction.** The sample begins with funds existing in 1984, so funds created later are not included.

- **Economic magnitude.** The top-decile alpha is statistically significant but small relative to the costs of frequent fund switching for taxable investors.
- **Daily data quality.** Daily NAV data create power but raise concerns about stale pricing and microstructure. The paper devotes substantial robustness analysis to this issue.
- **Interpretation of skill.** Persistence could reflect managerial ability, temporary information, exposure to hard-to-measure strategies, or other fund characteristics. The paper shows persistence, but does not fully decompose its source.
- **Short horizon only.** The evidence is intentionally short-run; the paper does not overturn the weak long-horizon performance evidence.

9 Detailed Table and Figure Walkthrough

9.1 Table 1: Risk-adjusted performance deciles: Ranking quarter

Purpose. Table 1 reports the ranking-quarter abnormal returns used to sort funds into deciles. It shows the cross-sectional spread in estimated stock selection, timing, and mixed strategy performance.

How to read it. Decile 1 is the highest-performing group in the ranking quarter; decile 10 is the lowest. The columns use different performance definitions: stock selection α_p , Treynor–Mazuy timing, Henriksson–Merton timing, and the two mixed methods.

Main takeaway. The ranking quarter contains very large estimated dispersion. In the stock-selection model, the top decile is 0.0738% per day and the bottom decile is -0.0828% per day. The dispersion is similar across the timing and mixed models.

9.2 Table 2: Risk-adjusted performance deciles: Post-ranking quarter

Purpose. Table 2 is the main persistence result. It measures the next-quarter abnormal performance of the funds sorted in Table 1.

Main takeaway. The top decile has significantly positive post-ranking abnormal performance under all major model specifications. In the stock-selection model, the top decile earns 0.0061% per day, or roughly 39 basis points per quarter. The bottom decile performs persistently poorly.

Table 1
Risk-adjusted performance deciles: Ranking quarter

Decile	Stock selection, α_p (%)	Market timing (%)		Mixed (%)	
		TM	HM	TM	HM
1	0.0738	0.0761	0.0768	0.0762	0.0754
2	0.0370	0.0376	0.0374	0.0376	0.0371
3	0.0221	0.0218	0.0223	0.0218	0.0221
4	0.0112	0.0108	0.0113	0.0108	0.0112
5	0.0017	0.0014	0.0017	0.0014	0.0017
6	-0.0072	-0.0078	-0.0077	-0.0078	-0.0073
7	-0.0163	-0.0171	-0.0174	-0.0171	-0.0165
8	-0.0275	-0.0287	-0.0291	-0.0287	-0.0280
9	-0.0426	-0.0442	-0.0451	-0.0441	-0.0437
10	-0.0828	-0.0839	-0.0870	-0.0838	-0.0844
Average	-0.0030	-0.0033	-0.0036	-0.0033	-0.0032

The table lists average daily performance estimates during the quarterly ranking period for deciles of funds sorted according to the performance estimates during the ranking period. We base the stock selection model performance rankings on α_p estimated from the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

We base the timing model rankings on the average $\alpha_p + \gamma_p f(r_{m,t})$ from the TM and HM market timing models,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \gamma_p f(r_{m,t}) + e_{p,t},$$

where $f(r_{m,t}) = r_{m,t}^2$ for TM, and $f(r_{m,t}) = I_t r_{m,t}$ for HM. The mixed rankings use the timing model $\alpha_p + \gamma_p f(r_{m,t})$ when the γ_p is statistically significant at the 5% level and otherwise use the α_p from the stock selection model. We assess the statistical significance of the γ_p using bootstrap standard errors. The table shows the average performance estimates during the quarterly ranking period. The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 2
Risk-adjusted performance deciles: Post-ranking quarter

Decile	Stock selection, α_p (%)	Market timing (%)		Mixed (%)	
		TM	HM	TM	HM
1	0.0061**	0.0040*	0.0039*	0.0040*	0.0056**
2	0.0006	0.0012	0.0014	0.0011	0.0011
3	-0.0003	0.0002	-0.0001	0.0004	-0.0020
4	-0.0001	-0.0002	-0.0011	-0.0003	0.0019
5	-0.0034**	-0.0023	-0.0041**	-0.0022	-0.0042
6	-0.0035**	-0.0042**	-0.0045**	-0.0042**	-0.0040**
7	-0.0054**	-0.0069**	-0.0046**	-0.0069**	-0.0065**
8	-0.0055**	-0.0056**	-0.0059**	-0.0056**	-0.0042**
9	-0.0075**	-0.0075**	-0.0079**	-0.0074**	-0.0080**
10	-0.0122**	-0.0123**	-0.0120**	-0.0125**	-0.0120**
Average	-0.0031	-0.0034	-0.0035	-0.0033	-0.0032

The table lists average daily performance estimates during the post-ranking period for deciles of funds sorted according to the performance estimates during the ranking period. We base the stock selection model performance rankings on α_p estimated from the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

We base the timing model rankings on the average $\alpha_p + \gamma_p f(r_{m,t})$ from the TM and HM market timing models,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \gamma_p f(r_{m,t}) + e_{p,t},$$

where $f(r_{m,t}) = r_{m,t}^2$ for TM, and $f(r_{m,t}) = I_t r_{m,t}$ for HM. The mixed rankings use the timing model $\alpha_p + \gamma_p f(r_{m,t})$ when the γ_p is statistically significant at the 5% level and otherwise use the α_p from the stock selection model. We assess the statistical significance of the γ_p using bootstrap standard errors. The table shows the average performance estimates during the following quarterly post-ranking period. * and ** indicate two-tailed significance at the 5 and 1% levels, respectively. The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 1: Risk-adjusted performance deciles: Ranking quarter.

Table 2: Risk-adjusted performance deciles: Post-ranking quarter.

9.3 Figure 1: Cumulative abnormal returns

Purpose. Figure 1 provides a visual version of the Table 2 stock-selection result by plotting cumulative abnormal returns for the top and bottom deciles during the post-ranking quarter.

Main takeaway. The top decile's cumulative abnormal return rises early in the quarter and remains positive. The bottom decile falls steadily. The figure makes clear that the persistence result is not just a single end-of-quarter observation.

9.4 Table 3: Post-ranking period performance deciles: Total returns

Purpose. Table 3 asks whether the same decile sorts produce noticeable differences in raw post-ranking total returns.

Main takeaway. Raw returns vary much less than risk-adjusted returns. The top and bottom stock-selection deciles have daily raw returns of 0.0585% and 0.0517%, respectively. The smaller raw-return spread shows why risk adjustment matters for identifying managerial ability.

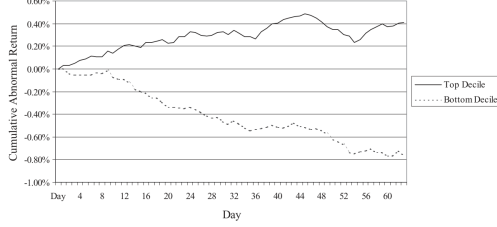


Figure 1
Cumulative abnormal returns
The figure shows cumulative average abnormal returns during a one-quarter post-ranking period for the top and bottom deciles of funds sorted according to α_p during a one-quarter ranking period. We estimate α_p with the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December

Figure 1: Cumulative abnormal returns.

29, 1995.

Table 3
Post-ranking period performance deciles: Total returns

Decile	Returns, R_p (%)	Stock selection, α_p (%)	Market timing (%)		Mixed (%)	
			TM	HM	TM	HM
1	0.0553	0.0585	0.0571	0.0568	0.0571	0.0578
2	0.0543	0.0538	0.0537	0.0549	0.0536	0.0539
3	0.0541	0.0541	0.0560	0.0549	0.0563	0.0527
4	0.0563	0.0555	0.0559	0.0544	0.0558	0.0579
5	0.0543	0.0565	0.0569	0.0553	0.0569	0.0557
6	0.0553	0.0543	0.0535	0.0522	0.0535	0.0526
7	0.0545	0.0534	0.0538	0.0545	0.0537	0.0523
8	0.0545	0.0550	0.0523	0.0546	0.0523	0.0576
9	0.0524	0.0520	0.0542	0.0548	0.0544	0.0522
10	0.0538	0.0517	0.0512	0.0524	0.0510	0.0520
Average	0.0545	0.0545	0.0545	0.0545	0.0545	0.0545

The table lists average daily total returns during quarterly post-ranking periods for deciles of funds sorted according to performance estimates during the quarterly ranking period. The first column represents rankings based on return, R_p . We base the stock selection model performance rankings on α_p estimated from the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

We base the timing model rankings on the average $\alpha_p + \gamma_p f(r_{m,t})$ from the TM and HM market timing models,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \gamma_p f(r_{m,t}) + e_{p,t},$$

where $f(r_{m,t}) = r_{m,t}^2$ for TM, and $f(r_{m,t}) = I_t r_{m,t}$ for HM. The mixed rankings use the timing model $\alpha_p + \gamma_p f(r_{m,t})$ when the γ_p is statistically significant at the 5% level and otherwise use the α_p from the stock selection model. We assess the statistical significance of the γ_p using bootstrap standard errors. This table shows the average daily total returns during the quarterly post-ranking period. The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 3: Post-ranking period performance deciles: Total returns.

9.5 Table 4: Post-ranking period performance deciles: Sharpe ratios

Purpose. Table 4 asks whether the post-ranking deciles differ in a simple return-per-unit-risk measure.

Main takeaway. Sharpe ratios show somewhat better risk-adjusted performance for the top abnormal-return decile, but the differences remain less dramatic than those in alpha. Again, factor-model risk adjustment carries important information beyond raw return and simple volatility scaling.

9.6 Table 5: Factor loadings of risk-adjusted performance deciles

Purpose. Table 5 explains why raw and risk-adjusted performance differ. It reports average factor loadings in the post-ranking period for deciles sorted by stock-selection alpha.

Main takeaway. The top and bottom deciles differ systematically in factor exposures. The top decile has relatively high size and momentum exposure and a more negative HML loading. Therefore, comparing raw returns alone understates the performance spread once differences in risk exposures are accounted for.

Table 4
Post-ranking period performance deciles: Sharpe ratios

Decile	Returns, R_p	Stock selection, α_p	Market timing		Mixed	
			TM	HM	TM	HM
1	0.0652	0.0707	0.0697	0.0685	0.0697	0.0701
2	0.0683	0.0680	0.0670	0.0687	0.0668	0.0676
3	0.0645	0.0648	0.0667	0.0673	0.0670	0.0637
4	0.0673	0.0679	0.0691	0.0647	0.0689	0.0714
5	0.0649	0.0690	0.0700	0.0683	0.0700	0.0665
6	0.0676	0.0656	0.0646	0.0636	0.0646	0.0645
7	0.0656	0.0635	0.0622	0.0652	0.0622	0.0625
8	0.0647	0.0626	0.0612	0.0621	0.0612	0.0649
9	0.0601	0.0611	0.0637	0.0633	0.0638	0.0618
10	0.0624	0.0574	0.0566	0.0592	0.0564	0.0578
Average	0.0651	0.0651	0.0651	0.0651	0.0651	0.0651

The table lists average daily Sharpe ratios during quarterly post-ranking periods for deciles of funds sorted according to performance estimates during the quarterly ranking period. The first column represents rankings based on return, R_p . We base the stock selection model performance rankings on α_p estimated from the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

We base the timing model rankings on the average $\alpha_p + \gamma_p f(r_{m,t})$ from the TM and HM market timing models,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \gamma_p f(r_{m,t}) + e_{p,t},$$

where $f(r_{m,t}) = r_{m,t}^2$ for TM, and $f(r_{m,t}) = I_{\gamma_p}(r_{m,t})$ for HM. The mixed rankings use the timing model $\alpha_p + \gamma_p f(r_{m,t})$ when the γ_p is statistically significant at the 5% level and otherwise use the α_p from the stock

application of gamma-macbeu (1975) inference is motivated by the potential for cross-fund correlation in the residuals of Equation (6) resulting from any systematic misspecification that would affect estimates of

Table 5
Factor loadings of risk-adjusted performance deciles

Decile	β_m	β_{unb}	β_{hml}	β_{mom}	R^2
1	1.045	0.274	-0.287	0.113	0.829
2	0.990	0.172	-0.189	0.065	0.859
3	0.968	0.137	-0.166	0.072	0.864
4	0.961	0.138	-0.127	0.059	0.871
5	0.961	0.117	-0.118	0.054	0.874
6	0.967	0.121	-0.114	0.047	0.872
7	0.982	0.145	-0.121	0.051	0.868
8	0.974	0.151	-0.115	0.026	0.863
9	1.006	0.203	-0.137	0.054	0.850
10	1.055	0.335	-0.200	0.044	0.805
Average	0.990	0.178	-0.157	0.058	0.856

The table lists average factor loadings estimated from the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t},$$

during quarterly post-ranking periods for deciles of funds sorted according to α_p during the ranking period. The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 4: Post-ranking period performance deciles: Sharpe ratios.

Table 5: Factor loadings of risk-adjusted performance deciles.

9.7 Table 6: Cross-sectional regression tests of performance persistence

Purpose. Table 6 estimates quarter-by-quarter cross-sectional persistence regressions of current performance on lagged performance.

Main takeaway. The slope coefficient B is positive and highly significant for risk-adjusted performance measures, but not for raw returns. This provides regression-based evidence that abnormal performance persists from one quarter to the next.

9.8 Table 7: Specification test

Purpose. Table 7 tests whether a simple size-momentum strategy, after expenses and transaction costs, can mechanically generate the abnormal returns found for the top mutual fund decile.

Main takeaway. Almost all simulated size-momentum portfolios have negative abnormal quarterly returns after costs. The strategies most comparable to the top mutual fund decile do not produce positive alphas. This weakens the explanation that the main result is just an unmodeled momentum strategy.

Table 6
Cross-sectional regression tests of performance persistence

	Returns, R_p (%)	Stock selection, α_p (%)	Market timing (%)		Mixed (%)	
			TM	HM	TM	HM
A	0.044	-0.002	-0.003	-0.003	-0.002	-0.002
p -value	.006	.213	.160	.126	.164	.181
B	0.036	0.122	0.118	0.117	0.118	0.117
p -value	.502	.000	.000	.000	.000	.000
R^2	0.101	0.038	0.034	0.032	0.034	0.036
Fraction of b coefficients						
Positive	.535	.767	.791	.721	.791	.767
p -value < .10	.395	.349	.372	.395	.372	.372
p -value < .05	.372	.349	.302	.349	.302	.326
p -value < .01	.349	.326	.302	.349	.302	.326
Negative	.465	.233	.209	.279	.209	.233
p -value < .10	.349	.047	.047	.070	.047	.047
p -value < .05	.349	.047	.047	.023	.047	.047
p -value < .01	.326	.047	.047	.023	.047	.047

The table shows results of cross-sectional regressions of fund performance during one quarterly period on fund performance during the previous quarterly period,

$$\text{Perf}_{p,t} = a + b\text{Perf}_{p,t-1} + e_{p,t}.$$

The first column represents rankings based on return, R_p . We base the stock selection model performance rankings on α_p estimated from the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

We base the timing model rankings on the average $\alpha_p + \gamma_p f(r_{m,t})$ from the TM and HM market timing models,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \gamma_p f(r_{m,t}) + e_{p,t}.$$

where $f(r_{m,t}) = r_{m,t}^2$ for TM, and $f(r_{m,t}) = I_{r_{m,t}}$ for HM. The mixed rankings use the timing model $\alpha_p + \gamma_p f(r_{m,t})$ when the γ_p is statistically significant at the 5% level and otherwise use the α_p from the stock selection model. We assess the statistical significance of the γ_p using bootstrap standard errors. We estimate a cross-sectional regression for each pair of successive quarterly periods. Listed first in the panels are the average regression parameters for the 43 quarterly regressions. The p -values are estimated in the spirit of Fama and MacBeth (1973) using the time series standard errors of the parameter estimates. Listed next are the fractions of slope coefficients that are positive (negative) and significant at the 10, 5, and 1% levels. The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 7
Specification test

Momentum quintile	Size quintile (%)				
	Small	2	3	4	Large
Low	-6.13	-4.02	-2.20	-1.00	0.74
2	-2.69	-1.82	-1.37	-0.98	0.96
3	-2.00	-1.59	-0.90	-0.78	-0.33
4	-1.68	-1.02	-1.02	-0.56	-0.16
High	-1.29	-0.76	-0.22	-0.50	-0.14

We sort stocks into quintiles quarterly on the basis of prior quarter return and market capitalization. We construct 25 simulated portfolios by value-weighting each stock that falls into a given intersection of the momentum and size sorts. We concatenate daily returns of the simulated portfolios over our sample period, and adjust to reflect expense ratios and transaction costs. Daily expense ratios are equivalent to 1.10% per year, the average annual expense ratio of the funds in our sample. We estimate transaction costs by size quintile using Keim and Madhavan (1998) and assume 100% turnover per year in the simulated portfolios. The table lists average abnormal quarterly returns based on α_p estimated from the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

The sample period is from January 2, 1985 to December 29, 1995.

Table 6: Cross-sectional regression tests of performance persistence.

Table 7: Specification test.

9.9 Table 8: Robustness test

Purpose. Table 8 repeats the post-ranking analysis using alternative momentum factors measured over 12 months, six months, three months, one month, one week, and one day.

Main takeaway. The top decile remains positive and statistically significant across all momentum-factor definitions. The evidence is therefore not driven by one particular implementation of the momentum control.

9.10 Table 9: Top decile post-ranking period performance by length of ranking and post-ranking period: Time series averages – sorting on abnormal return

Purpose. Table 9 asks what happens when the ranking and post-ranking windows are lengthened from one quarter to one year and three years, while sorting on abnormal return.

Main takeaway. The quarterly daily-data result is strong: 0.385% per quarter for the top decile, with a t-statistic of 3.165. The one-year and three-year estimates are small and insignificant. Persistence is visible only at short horizons.

before, both for ranking and holding purposes.

Table 8 lists the results. In all cases, the top decile generates significant, positive abnormal returns in the post-ranking quarter. The

Table 8
Robustness test

Decile	12-mo (%)	<i>t</i> -stat	6-mo (%)	<i>t</i> -stat	3-mo (%)	<i>t</i> -stat	1-mo (%)	<i>t</i> -stat	1-wk (%)	<i>t</i> -stat	1-day (%)	<i>t</i> -stat
1	0.0061	3.17	0.0085	4.11	0.0084	3.86	0.0107	5.08	0.0070	2.89	0.0064	2.80
2	0.0006	0.40	0.0038	2.50	0.0038	2.49	0.0056	3.66	0.0045	2.38	0.0037	2.17
3	-0.0003	-0.24	0.0005	0.34	0.0019	1.21	0.0014	0.89	0.0036	1.98	0.0012	0.76
4	-0.0001	-0.09	-0.0006	-0.43	0.0021	1.37	0.0021	1.55	-0.0014	-0.88	-0.0009	-0.54
5	-0.0034	-2.69	-0.0015	-1.13	-0.0006	-0.44	0.0022	1.65	0.0029	1.79	-0.0039	-2.67
6	-0.0035	-2.63	-0.0029	-2.07	-0.0020	-1.52	-0.0031	-2.18	-0.0015	-0.95	-0.0013	-0.89
7	-0.0054	-3.94	-0.0047	-3.50	-0.0023	-1.68	-0.0025	-1.71	-0.0019	-1.31	-0.0013	-0.88
8	-0.0055	-3.92	-0.0056	-3.97	-0.0050	-3.50	-0.0033	-2.36	-0.0042	-2.44	-0.0030	-2.01
9	-0.0075	-4.76	-0.0057	-3.85	-0.0041	-2.74	-0.0054	-3.77	-0.0033	-1.83	-0.0068	-4.16
10	-0.0122	-6.22	-0.0122	-6.24	-0.0110	-5.68	-0.0118	-6.04	-0.0114	-5.11	-0.0109	-5.08

The table lists average daily performance estimates during quarterly post-ranking periods for deciles of funds sorted according to α_p during quarterly ranking periods. We estimate α_p with the four-factor model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

The columns show the results for alternative momentum factors. All are value-weighted indices created by subtracting the return of 30% of stocks with the lowest returns from the return of 30% of stocks and the highest returns as measured over horizons ranging from 12 months to 1 day. We re-form the factors constructed from horizons of 1 month or greater each month. We re-form the weekly factor weekly. We re-form the daily factor daily. The sample period is from January 2, 1985 to December 29, 1995.

Table 9
Top decile post-ranking period performance by length of ranking and post-ranking period: Time series averages — sorting on abnormal return

Measurement interval	Data frequency		
	Daily	Weekly	Monthly
Quarterly	0.385% (3.165)		
1-yr	0.070% (0.514)	0.034% (0.238)	
3-yr	0.049% (0.518)	-0.133% (-1.382)	-0.117% (-1.002)

The table lists average abnormal quarterly returns during post-ranking periods for the top decile of funds sorted by abnormal return during ranking periods. We measure abnormal returns by α_p estimated from the four-factor stock selection model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + e_{p,t}.$$

We estimate post-ranking abnormal returns separately over each post-ranking period. For the quarterly measurement intervals, we rank each quarter. For the one- and three-year measurement intervals, we rank each year. In this table, the ranking and post-ranking periods are the same duration (i.e., one quarter, one year, or three years). Listed in parentheses below each average abnormal quarterly return are *t*-statistics estimated in the spirit of Fama and MacBeth (1973) using the time series standard errors of the parameter

Table 8: Robustness test with alternative momentum factors.

Table 9: Time series averages – sorting on abnormal return.

9.11 Table 10: Top decile post-ranking period performance by length of ranking and post-ranking period: Time series averages – sorting on return

Purpose. Table 10 repeats Table 9 but sorts funds on raw return rather than abnormal return.

Main takeaway. Sorting on raw return does not identify a superior top decile. The quarterly estimate is only 0.156% and statistically insignificant. This result is central to the paper’s reconciliation with prior work that relied on raw-return rankings.

9.12 Table 11: Top decile post-ranking period performance by length of ranking and post-ranking period: Concatenated series – sorting on abnormal return

Purpose. Table 11 changes the performance evaluation method. Instead of estimating post-ranking performance separately in each period and averaging, it concatenates post-ranking returns and estimates alpha once over the whole series.

Main takeaway. The abnormal-return sort no longer produces significant alpha. This shows that the evidence depends on a conditional, quarter-by-quarter evaluation rather than a single unconditional long time-series estimate.

estimated in the spirit of Fama and MacBeth (1973) using the time series standard errors of the parameter estimates. The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 10
Top decile post-ranking period performance by length of ranking and post-ranking period: Time series averages—sorting on return

Measurement interval	Data frequency		
	Daily	Weekly	Monthly
Quarterly	0.156% (1.180)		
1-yr	-0.032% (-0.259)	-0.069% (-0.521)	
3-yr	-0.102% (-1.045)	-0.185% (-1.986)	-0.056% (-0.508)

The table lists average abnormal quarterly returns during post-ranking periods for the top decile of funds sorted by return during ranking periods. We measure abnormal returns by α_p estimated from the four-factor stock selection model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \varepsilon_{p,t}.$$

We estimate post-ranking abnormal returns separately over each post-ranking period. For the quarterly measurement intervals, we rank each quarter. For the one- and three-year measurement intervals, we rank each year. In this table, the ranking and post-ranking periods are the same duration (i.e., one quarter, one year, or three years). Listed in parentheses below each average abnormal quarterly return are t -statistics estimated in the spirit of Fama and MacBeth (1973) using the time series standard errors of the parameter estimates. The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

here. We interpret these results as evidence that superior performance is a short-lived phenomenon that can only be detected using relatively short

Table 10: Time series averages – sorting on return.

Table 11
Top decile post-ranking period performance by length of ranking and post-ranking period: Concatenated series—sorting on abnormal return

Measurement interval	Data frequency		
	Daily	Weekly	Monthly
Quarterly	0.092% (0.261)		
1-yr	0.336% (0.974)	0.339% (0.998)	
3-yr	-0.048% (-0.157)	-0.228% (-0.751)	-0.347% (-1.047)

The table lists average abnormal quarterly returns during post-ranking periods for the top decile of funds sorted by abnormal return during ranking periods. We measure abnormal returns by α_p estimated from the four-factor stock selection model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \varepsilon_{p,t}.$$

We estimate post-ranking abnormal returns once over the entire sample using a concatenated post-ranking return series. For the quarterly measurement intervals, we rank each quarter. For the one- and three-year measurement intervals, we rank each year. For the one-quarter and one-year measurement intervals in this table, the ranking and post-ranking periods are the same duration (i.e., one quarter, one year, or three years). The three-year rows correspond to a three-year ranking period and a one-year post-ranking period, which gives a single, continuous, concatenated, post-ranking period return series. Here,

Table 11: Concatenated series – sorting on abnormal return.

9.13 Table 12: Top decile post-ranking period performance by length of ranking and post-ranking period: Concatenated series – sorting on return

Purpose. Table 12 combines two Carhart-like choices: sorting on raw return and evaluating with a concatenated post-ranking series.

Main takeaway. No significant positive performance remains. The quarterly estimate is negative, and one-year and three-year estimates are insignificant. This table is the most direct demonstration that moving toward the standard long-horizon methodology eliminates the short-term persistence result.

statistics are standard OLS (listed in parentheses). The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 12
Top decile post-ranking period performance by length of ranking
and post-ranking period: Concatenated series — sorting on return

Measurement interval	Stock selection, α_p
Quarterly	−0.279% (−0.659)
1-yr	0.139% (0.405)
3-yr	−0.487% (−1.357)

The table lists average abnormal quarterly returns during post-ranking periods for the top decile of funds sorted by return during ranking periods. We measure abnormal returns by α_p estimated from the four-factor stock selection model,

$$r_{p,t} = \alpha_p + \sum_{k=1}^4 \beta_{pk} r_{k,t} + \varepsilon_{p,t}.$$

We estimate post-ranking abnormal returns once over the entire sample using a concatenated post-ranking monthly return series. For the quarterly measurement intervals, we rank each quarter. For the one- and three-year measurement intervals, we rank each year. For the one-quarter and one-year measurement intervals in this table, the ranking and post-ranking periods are the same duration (i.e., one quarter, one year, or three years). The three-year rows correspond to a three-year ranking period and a one-year post-ranking period, which gives a single, continuous, concatenated, post-ranking period return series. Here, the t-statistics are standard OLS (listed in parentheses). The sample consists of 230 mutual funds. The sample period is from January 2, 1985 to December 29, 1995.

Table 12: Concatenated series – sorting on return.

10 Bottom Line

Bollen and Busse show that mutual fund performance persistence exists, but it is short-lived. Daily data allow them to estimate quarterly risk-adjusted performance, and the top decile continues to earn positive abnormal returns in the following quarter. The result is robust to alternative performance models and momentum controls, but it disappears when the researcher ranks on raw returns, lengthens the measurement horizon, or imposes a single concatenated long time-series evaluation. The paper’s broad lesson is methodological and economic: mutual fund ability may be conditional and temporary, and high-frequency short-window tests can reveal patterns that long-horizon tests average away.